A Lean-verified reduction of periodic Navier–Stokes regularity to a single integrability hypothesis, with a tethered-model ceiling

(conditional results; draft)

Benjamin Stanley Frohman
ORCID 0009-0006-7068-3718

Draft, October 31, 2026 — private, not peer reviewed, no DOI

Scope. This paper reports *conditional* results. It does **not** solve the Clay Millennium Problem. It does not prove global regularity of the 3D Navier–Stokes equations, either periodic or on $\mathbb{R}^3$. The single remaining hypothesis of Part I, `PeriodicBKMIntegrability`, is equivalent in difficulty to Clay Statement (B). Part II concerns a *modified* (tethered) model, not Navier–Stokes, and is itself conditional.

Abstract

We describe two machine-checked Lean 4/Mathlib developments. Part I, on the torus $\mathbb{T}^3$ (repository `periodic-ns-classical-continuation`, commit `4ab624e`), proves the following with no hypothesis Props:

- H^3 Beale–Kato–Majda control, `periodicBKMHsControl_three`;
- the periodic BKM blow-up criterion, `periodicBKMCriterion`.

It deduces the exact periodic Clay Statement (B), `Clay_B_exact`, conditionally on a single named Prop, `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). That Prop asserts $\int_0^T \|\omega\|_\infty \, dt < \infty$ along maximal solutions. By the proved criterion it is equivalent to Statement (B), so it is exactly as hard.

Part II (repository `metriplectic-world-model`, commit `2274a23`) proves a uniform-in-time ceiling on $\sup|\omega|$ for a vorticity equation with a cubic damping tether, `field_uniform_ceiling_logLoss_cubicTether`. It is conditional on a pointwise log-loss stretching Prop, `FieldStretchingLogLoss`, and on fixed compact support and regularity hypotheses, which we list. We explain why the known Calderón–Zygmund log estimate does not supply that Prop. All results use only the axioms `propext`, `Classical.choice` and `Quot.sound`.

1 Part I: the periodic chain on $\mathbb{T}^3$

Throughout, $\nu > 0$, u_0 is smooth, periodic and divergence-free, and (u, p) is the maximal smooth solution with existence time T^*. The definitions `Clay_B_exact`, `PeriodicBKMCriterion` and `PeriodicBKMIntegrability` come from the dependency `fefferman-clay-navier-stokes`.

Hypothesis 1 (`PeriodicBKMIntegrability` ν; **open**)**.** For every maximal solution and every T, $t \mapsto \|\omega(t)\|_\infty$ is integrable on $[0, T]$.

Theorem 1 (`periodicBKMHsControl_three`†; unconditional)**.** *For every ν, `PeriodicBKMHsControl` ν 3 holds. That is, if $T^* = T < \infty$ and $\int_0^T \|\omega\|_\infty < \infty$, then $\sup_{[0,T)} \|u(t)\|_{H^3} < \infty$.*

Proof route (all formalized). The H^3 energy identity reduces the claim to a commutator bound and a low-frequency vorticity bound.

- The $s = 3$ commutator bound is $-2\sum_i \int (1-\Delta/4\pi^2)^3 v_i\,((v\cdot\nabla)v)_i \le C\|\nabla v\|_\infty \|v\|_{H^3}^2$ (`periodicPhysicalComm` `CommutatorAssembly.lean`).
- The low-frequency bound is $\|V_R \nabla v\|_\infty \le K(1+\log R)\|\omega\|_\infty$ for $R \ge 2$ (`periodicLowFreqVorticityBound`†, `LowFreqAssembly.lean`). It is proved through the Riesz form of Biot–Savart, a smooth dyadic decomposition with uniformly L^1-bounded block kernels, and a uniform L^1 bound for the de la Vallée Poussin kernel.

Grönwall's inequality with the log splitting then closes the bound.

Theorem 2 (`periodicBKMCriterion`†; unconditional)**.** *For every ν, `PeriodicBKMCriterion` ν holds. If $\nu > 0$ and $T^* = T < \infty$, then $\int_0^T \|\omega\|_\infty = \infty$.*

Theorem 3 (`clay_B_exact_of_integrability`†; conditional)**.** *`PeriodicBKMIntegrability` $\nu \Rightarrow$ `Clay_B_exact` ν.*

Remark 1 (Equivalence in difficulty)**.** The converse direction also holds. If Statement (B) gives a global smooth solution, then by uniqueness the maximal solution is global and smooth, so $\|\omega\|_\infty$ is continuous and integrable on every $[0, T]$. Hence `PeriodicBKMIntegrability` is *equivalent* to the periodic Clay Statement (B); it is not a weaker or "technical" leftover. This converse is argued informally here and is not packaged as a separate Lean theorem. Part I is therefore a verified reduction, not a proof of regularity.

2 Part II: a ceiling for the cubic-tethered vorticity model

The model is `ModelVorticityPDE` $\nu\ \omega\ u\ T$:

$$u = \mathrm{BS}(\omega), \qquad \operatorname{div}\omega = 0, \qquad \partial_t\omega + (u\cdot\nabla)\omega - (\omega\cdot\nabla)u = \nu\Delta\omega + T.$$

It uses the cubic tether $T =$ `cubicTether` $\kappa\ \omega = -\kappa|\omega|^2\omega$. This is *not* Navier–Stokes; the tether is an added damping term. Write $Y(t) = \sup_x |\omega(t,x)|$ and $\mathrm{ceiling}(M_0, C, \kappa) = \max(M_0, \max(C/\kappa, 0))$.

Hypothesis 2 (`FieldStretchingLogLoss` $\omega\ u\ Y\ C_1\ C_2$†; **not proved**)**.** For every $t \ge 0$ and every global maximizer x_* of $|\omega(t,\cdot)|$:

$$\omega\cdot(\omega\cdot\nabla)u\,(t,x_*) \le C_1 Y^3 + C_2 Y^3\log(1+Y).$$

Theorem 4 (`field_uniform_ceiling_logLoss_cubicTether`†; conditional)**.** *Assume all of the following:*

1. *$\nu \ge 0$, $\kappa > 0$, $C_2 \ge 0$;*
2. *`ModelVorticityPDE` $\nu\ \omega\ u$ (`cubicTether` $\kappa\ \omega$);*
3. *a* fixed *compact nonempty set K with $\omega(s,x) = 0$ for $x \notin K$, for all s;*
4. *$\omega(t) \in C^2$ in space for every t;*
5. *joint continuity of ω and of $\partial_t\omega$, and the pointwise time derivative $\partial_t\omega(s,x)$ exists for all s, x;*

6. *$Y(s)$ is the least upper bound of $\{|\omega(s,x)|\}$ for all s;*

7. *`FieldStretchingLogLoss` ω u Y C_1 C_2.*

Then for all $t \geq 0$, $Y(t) \leq \mathrm{ceiling}\big(Y(0),\, C_1 + \kappa/4 + 4C_2^2/\kappa,\, 3\kappa/4\big)$.

Proof route.

- `fieldNetRate_of_logLoss_cubicTether`† uses $\log(1+y) \leq 2\sqrt{1+y}$ and Young's inequality to absorb the log loss into the quartic tether. This gives `FieldNetRate` with $C = C_1 + \kappa/4 + 4C_2^2/\kappa$ and $\kappa_{\mathrm{eff}} = 3\kappa/4$.
- `field_uniform_ceiling_of_netRate`† applies a Danskin upper-Dini bound for Y^2 at a maximizer. There the transport term vanishes and $\omega \cdot \Delta\omega \leq 0$.
- A Dini comparison lemma, `ceiling_of_upper_dini`†, finishes the proof. No differentiability of Y is assumed.

Why the known Calderón–Zygmund estimate does not suffice. The classical log estimate (Beale–Kato–Majda, Kozono–Taniuchi type) reads

$$\|\nabla u\|_\infty \leq C\big(1 + Y\,(1 + \log^+(\|\omega\|_{C^\alpha}/Y)) + \|\omega\|_{L^2}\big).$$

The logarithm contains a *higher* norm, $\|\omega\|_{C^\alpha}$. The ceiling argument controls only Y, and a time-uniform bound on $\|\omega(t)\|_{C^\alpha}$ is essentially the regularity question itself. So the known estimate does not imply `FieldStretchingLogLoss`, whose Y-only form is a strictly stronger hypothesis than anything known.

The CZ estimate itself is also not formalized in that repository, which has no singular-integral or Hölder theory for ∇BS. The existence of solutions with uniformly fixed compact support (hypothesis 3) is likewise not established. Neither the theorem nor anything else in Part II says anything about unmodified Navier–Stokes.

3 Status

- **Proved, no hypothesis Props:** `periodicBKMHsControl_three`, `periodicBKMCriterion`, `periodicLowFreqVorticityBound` and `periodicPhysicalCommutatorBound`. In Part II: `fieldNetRate_of_logLoss_cubicTether`, an implication between named Props.
- **Conditional:** `clay_B_exact_of_integrability`, on `PeriodicBKMIntegrability`. `field_uniform_ceili` on `FieldStretchingLogLoss` and hypotheses 1–6.
- **Open:** `PeriodicBKMIntegrability`, which is equivalent to Clay (B), and `FieldStretchingLogLoss`. Neither the Clay problem nor Navier–Stokes regularity is claimed.

A Theorem-to-Lean map

Names marked † were coined during formalization and are not the author's terminology; they may be renamed later.

Lean name	File	Repo @ commit
`periodicBKMHsControl_three`†	`PeriodicNSContinuation/BKMFinal.lean`	pnscc @ 4ab624e
`periodicBKMCriterion`†	`PeriodicNSContinuation/BKMFinal.lean`	pnscc @ 4ab624e
`clay_B_exact_of_integrability`†	`PeriodicNSContinuation/BKMFinal.lean`	pnscc @ 4ab624e
`periodicBKMHsControl_three_of_lowFreq`†	`PeriodicNSContinuation/CommutatorAssembly.lean`	pnscc @ 4ab624e
`periodicPhysicalCommutatorBound`†	`PeriodicNSContinuation/CommutatorAssembly.lean`	pnscc @ 4ab624e
`periodicLowFreqVorticityBound`†	`PeriodicNSContinuation/LowFreqAssembly.lean`	pnscc @ 4ab624e
`periodicBKMCriterion_of_hsControl`†	`PeriodicNSContinuation/Continuation.lean`	pnscc @ 4ab624e
`PeriodicBKMHsControl`† (def)	`PeriodicNSContinuation/Continuation.lean`	pnscc @ 4ab624e
`clay_B_exact_of_bkm`†	`PeriodicNSContinuation/LocalExistence.lean`	pnscc @ 4ab624e
`Clay_B_exact` (def)	`FeffermanClayNS/FeffermanStatement.lean`	pnscc @ 4ab624e
`PeriodicBKMCriterion`, `PeriodicBKMIntegrability` (defs)	`FeffermanClayNS/HighSetDepletionAndBKM.lean`	pnscc @ 4ab624e
`field_uniform_ceiling_logLoss_cubicTether`†	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ 2274a23
`fieldNetRate_of_logLoss_cubicTether`†	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ 2274a23
`field_uniform_ceiling_of_netRate`†	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ 2274a23
`FieldStretchingLogLoss`†, `FieldNetRate`† (defs)	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ 2274a23
`cubicTether`†, `fieldTetherFeedback_cubicTether`†	`MetripelecticWorldModel/DiniCeiling.lean`	mwm @ 2274a23
`ceiling` (def)	`MetripelecticWorldModel/Ceiling.lean`	mwm @ 2274a23
`ModelVorticityPDE`† (def)	`MetripelecticWorldModel/SmoothOrbit.lean`	mwm @ 2274a23

pnscc = `BenFrohman/periodic-ns-classical-continuation`; mwm = `metriplectic-world-model`.

B Axiom audit

In each repository, `lake build` succeeds with 0 `sorry`, and `lake env lean AxiomAudit.lean` passes. Every audited theorem, including all those listed above, depends only on `propext`, `Classical.choice` and `Quot.sound`. In pnscc, the audit prints "AxiomAudit PASS: only propext, Classical.choice, Quot.sound". In mwm, the audit throws an error if any theorem uses another axiom, and it completed without error.

License

Copyright 2026 Benjamin Stanley Frohman. Apache License, Version 2.0.